Consistency-Preserving On-Device PII Substitution with a 1-bit Small Language Model

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Abstract

PII redaction in machine-learning pipelines today usually replaces detected entities with placeholder tokens such as `[PERSON]`, destroying the downstream utility of the redacted text for retrieval, language-model fine-tuning, and NER training. We propose and evaluate a fully on-device pipeline that *substitutes* PII with consistent, type-preserving fake values: a 1.5 B mixture-of-experts token classifier (`openai/privacy-filter`, with 50 M active parameters) detects spans, an entity resolver groups same-string mentions, and a 1-bit small language model (`Bonsai-1.7B`, Q1_0) proposes contextual surrogates for names, addresses, and dates while a rule-based generator (`faker`) handles fully patterned fields. We are not aware of a prior published evaluation of an end-to-end on-device PII substitution stack that combines a token-classifier, a 1-bit SLM, and a rule-based generator; this paper aims to fill that gap. We report a *prompting* finding that turns out to be more important than the quantization choice: with a naive fixed-three-shot demonstration, both 1-bit `Bonsai-1.7B` and 1.58-bit `Ternary-Bonsai-1.7B` regurgitate the demonstration outputs verbatim regardless of the input—the Chinese name 杨娟, the Japanese name 山田太郎, and the German name Müller-Schulz all collapse to “Alice Johnson”. We fix this with **locale-conditioned rotating few-shot demonstrations**: a character-range heuristic classifies each input’s locale and selects a locale-pure demonstration pool, and a per-input MD5 hash deterministically samples which three demonstrations appear in the prompt. With the fix, the same 1-bit model produces locale-correct surrogates (杨娟→李伟, Müller-Schulz→Anna Becker, `11-Jul-1998`→`05-Aug-2003`), achieves downstream NER training F1 within seed-level noise of faker-only baselines (0.371 ± 0.049 vs. 0.448 ± 0.082, $n = 5$ seeds, Welch’s $p \approx 0.12$), and beats faker on length preservation in 5 of 6 locales. We argue that small-LM PII substitution is viable on-device, and that prompt engineering matters more than quantization at this scale. The code repository for this work is accessible at `https://github.com/asadani/on-device-pii-substitution`.

1 Introduction

Existing PII redaction tools—Microsoft Presidio [7], the OpenAI privacy filter [9], `spaCy`+regex pipelines [4]—produce *redacted* text in which detected entities are replaced by short placeholder tokens such as `[NAME]` or `[EMAIL]`. This is adequate for compliance display purposes but fails for downstream uses that need natural-looking text: training data for fine-tuning, search indices over redacted corpora, and retrieval-augmented generation over sensitive document stores. The placeholder approach also hurts utility in measurable ways: NER models trained on `[PERSON]`-redacted corpora generalise poorly to non-redacted text (Section 4.5), and language-model perplexity over placeholder-rich documents is dominated by the placeholder tokens themselves.

Substitution—replacing each detected PII span with a realistic, fake value of the same type—is the obvious alternative, but three constraints have so far prevented its widespread adoption in privacy-sensitive pipelines: **(1)** the substitution must be *consistent* within a document, so that "John Smith" → "Marcus Chen" everywhere, not five different fakes; **(2)** it must be *type-preserving*, so a Chinese name does not become a US name; and **(3)** it should run *on-device* without sending the (still-real-PII) input to a cloud LLM [8].

Recent advances in 1-bit and ternary-bit quantization (e.g., the Bonsai [10] and Ternary-Bonsai [11] families, and prior work on BitNet [6]) make small generative language models viable on commodity CPUs. We investigate whether such models can serve as the surrogate-proposer in a privacy-preserving substitution pipeline, evaluating the resulting system on five metrics (privacy, naturalness, consistency, length, and downstream NER training F1).

Threat model (scope). We measure literal-string leak against ground-truth PII values; we do not consider membership-inference or linkage attacks against the substituted output. Differential-privacy guarantees on the substitution distribution are out of scope.

Contributions.

1. A reproducible on-device substitution pipeline combining `openai/privacy-filter`, `Bonsai-1.7B` (Q1_0), and `faker`, with all code and configurations released.
2. A quantitative comparison of redact / faker-only / hybrid substitution modes under the constraint of CPU-only inference, on 100 multi-template multilingual documents. We are not aware of a directly comparable prior result that fixes all three constraints (consistency, type-preservation, and on-device execution) together.
3. **Identification and isolation of a "few-shot regurgitation" failure mode** in small SLMs at extreme quantization, validated by showing that 1-bit `Bonsai-1.7B` and 1.58-bit `Ternary-Bonsai-1.7B` produce identical regurgitating outputs on the same inputs (i.e., the failure is caused by prompting, not by quantization).
4. **A simple and effective fix: locale-conditioned rotating few-shot demonstrations**, with deterministic per-input demo sampling that preserves cache-friendliness. The fix eliminates the regression in downstream NER F1 that we observed under naive prompting.

2 Related Work

2.1 PII detection

Microsoft Presidio [7] combines regex, `spaCy` NER, and rule-based recognisers. The `openai/privacy-filter` model [9] (1.5 B MoE, 50 M active) is a fine-tuned encoder using BIOES / BILOU-style token tags (popularised in NER by Ratinov and Roth [13]) over eight PII categories. Running the inference harness shipped with the model on its bundled seven-template synthetic benchmark (`samples.json`, N=100), we measure F1 = 0.587 (precision 0.619, recall 0.627), with the largest source of false positives being dates (44.5% of all FPs) and the largest source of false negatives being non-English addresses (these numbers are computed by us, from the same harness; see Section 4.1). We use these as the detection-side floor for all subsequent comparisons, since every mode in our experiments inherits the same detector and therefore the same recall ceiling.

2.2 Synthetic PII and anonymization

`faker` [2] is the de-facto rule-based generator for synthetic PII. Presidio's anonymizer can swap detected spans for hash, redaction, or counter-style replacements but not consistent type-preserving fakes. Recent work has used GPT-3.5 / GPT-4 [8] to generate context-appropriate replacements; this approach violates the on-device constraint and incurs per-request cost.

2.3 Low-bit small language models

The Bonsai family (1-bit Q1_0) [10] and Ternary-Bonsai (1.58-bit Q2_0) [11] are Qwen3-based [12] decoder-only LMs trained for extreme quantization. We run inference via `llama.cpp` [3]. Prior work [6] has evaluated extreme-quantization SLMs on QA-style benchmarks; we did not find published evaluations of these models on generative-substitution tasks where input copying or demonstration-regurgitation is a failure mode (the specific failure we document in Section 4.3).

2.4 Downstream-utility evaluation

Prior anonymization work typically reports privacy and naturalness in isolation. We follow the broader synthetic-data literature in additionally measuring length preservation and within-document entity consistency, both relevant to downstream NER training utility, and we evaluate the latter directly by training a `spaCy` [4] NER model on each variant of our corpus and testing on held-out original text.

3 Method

3.1 Architecture

```
text -> privacy-filter (BIOES -> char spans)
     -> EntityResolver (group by canonical, label)
     -> propose_surrogate dispatcher:
         PERSON / ADDRESS / DATE -> Bonsai-1.7B
         EMAIL / PHONE / ACCT / URL / SECRET -> faker
     -> splice (R-to-L, preserve whitespace)
     -> output text
```

Detection runs once per document at ∼1 s/512 tokens on CPU. The `EntityResolver` groups detected spans by `(canonical_lowercased, label)` so that all mentions of “John Smith” share a single surrogate. Surrogates are cached per `(mode, family, canonical, label)` so repeated names and common patterns trigger the SLM exactly once over the corpus.

3.2 Surrogate proposers

For PERSON, ADDRESS, and DATE labels, we invoke `Bonsai-1.7B` (Q1_0) through `llama-cli -single-turn`. Each call uses a three-shot prompt of the form `Real: <example>\nFake: <example>\n...Real: <input>\nFake:`.

Locale-conditioned rotating demonstrations. Naively using a small fixed set of demonstrations is catastrophic for 1-bit SLMs: the model pattern-matches the demonstration *outputs* and emits one of them verbatim regardless of the input. A pilot run of the pipeline with three fixed English demonstrations produced “Alice Johnson” as the surrogate for the Chinese name 杨娟, the Japanese name 山田太郎, and the German name Müller-Schulz—all collapsed to the first English demonstration value (see Section 4.3 for the complete diagnosis).

Our final design avoids this with two cooperating mechanisms:

1. **Locale conditioning.** A lightweight character-range and keyword-based heuristic classifies each input into one of {`en`, `de`, `es`, `ja`, `zh`} for PERSON / ADDRESS, and one of {`mdy_slash`, `ymd_dash`, `dmy_dash_mon`, `dmy_slash`, `unknown`} for DATE. Each class has its own pool of 4–8 demonstrations in the matching script and date format (Appendix A).
2. **Per-input rotating sampling.** From the appropriate pool, three demonstrations are sampled deterministically with an MD5 hash of the input string as the random seed. This means *the*

same entity always receives the same demonstrations (cache-friendly across documents) but *different entities receive different demonstration subsets* (preventing any single demonstration value from dominating the surrogate distribution).

With this strategy, `Bonsai-1.7B` produces locale-appropriate, format-preserving, and diverse surrogates: 杨娟→李伟, 山田太郎→郑强 (Chinese fallback for kanji-only Japanese names—see Section 5), Müller-Schulz→Anna Becker, `11-Jul-1998`→`05-Aug-2003`.

We validate every response and reject empty, identity-equal, or punctuation-only outputs, falling back to `faker` when validation fails.

For EMAIL, PHONE, ACCT, URL, and SECRET labels, we use `faker` directly because these fields have low contextual entropy—there is no benefit from generative reasoning over `Faker().email()`.

3.3 Splice and whitespace

The privacy filter sometimes includes a leading whitespace in its character span. We preserve original leading and trailing whitespace when splicing surrogates back into the text to avoid cosmetic artefacts such as `is[PRIVATE_PERSON]` (which would inflate perplexity).

4 Experiments

4.1 Setup

Dataset. 100 documents from the `openai/privacy-filter` evaluation set [1, 9], covering 7 templates (1099, W-2, auto_insurance, bank_statement, invoice, mortgage_insurance, paystub) and 6 locales (en_US n=42, en_IN 16, de_DE 12, es_MX 10, ja_JP 10, zh_CN 10). Each document carries ground-truth PII values (∼6.6 per document on average).

Models.

- Detection: `openai/privacy-filter`, 1.5 B MoE, 50 M active, bf16, CPU [9, 14].
- SLM: `Bonsai-1.7B` Q1_0 [10], CPU via `llama.cpp` [3].
- Naturalness evaluator: `distilgpt2` (∼82 M parameters) [5], CPU.
- Fallback surrogate generator: `faker` [2].
- Downstream NER: `spaCy` blank English pipeline [4].

Hardware. 31 GB RAM, x86-64 CPU; no GPU used.

Modes evaluated.

- `redact`: replace each span with `[LABEL]` (current state of practice).
- `faker`: replace every span with a `faker` value of matching type.
- `hybrid`: `Bonsai-1.7B` for PERSON / ADDRESS / DATE, `faker` for the other five labels (our proposed pipeline).

4.2 Primary metrics

We report four per-document metrics, averaged across the corpus:

1. **Leak rate**: fraction of *ground-truth PII strings* that still appear verbatim (case-insensitive substring) in the output. Lower is better.

Table 1: Primary metrics (N=100, averaged across all 7 templates and 6 locales).

Mode	Leak ↓	PPL ↓	Consistency ↑	Length pres. ↑	Avg. latency
redact	0.226	27.8	1.000	0.973	1.2 s
faker	0.226	47.3	1.000	0.978	1.1 s
hybrid	0.226	**42.8**	1.000	**0.982**	38.1 s

2. **Naturalness PPL**: `distilgpt2` perplexity over the output text, chunked at 1024 tokens. Lower is more natural.
3. **Consistency rate**: for entities appearing ≥ 2 times in the input, fraction where the output uses the same surrogate at every mention. Higher is better.
4. **Length preservation**: $1 - |\,\mathrm{len(out)} - \mathrm{len(in)}\,|/\mathrm{len(in)}$. Closer to 1 is better.

Headline result. Hybrid PPL (42.8) is lower than faker (47.3, −9.5%) while length preservation is the best of the three modes (0.982 vs. 0.978, 0.973). Redact achieves the lowest PPL (27.8), but this is an artefact of `distilgpt2`'s tokenisation (placeholder tokens cluster predictably—not because the text is more natural; see Section 5). Consistency is 1.000 across all modes because surrogates are deterministically cached per `(mode, family, canonical, label)`.

Privacy floor. The leak rate of 0.226 is *identical* across all three modes: every mode misses the same ground-truth values, consistent with the detection-side recall (≈ 0.627) we measured for the privacy filter on the same data (Section 2). The architectural choice between redact / faker / hybrid does not affect privacy at all—only *what to do* with the spans the filter catches.

Per-locale (PPL). Hybrid is the PPL winner over faker in **all six locales**: en_US (42.2 vs. 46.4), en_IN (38.5 vs. 44.5), de_DE (47.6 vs. 51.3), es_MX (32.6 vs. 35.6), ja_JP (44.1 vs. 49.7), zh_CN (55.3 vs. 59.9). The English-only `distilgpt2` proxy systematically penalises non-Latin script outputs, so the *true* multilingual naturalness gap is likely larger than these numbers suggest.

Per-locale (length preservation). Hybrid is the length-preservation winner over faker in **five of six locales** (de_DE 0.984 vs. 0.975, en_IN 0.982 vs. 0.972, es_MX 0.986 vs. 0.980, zh_CN 0.995 vs. 0.971, ja_JP 0.974 vs. 0.976 near-tie); only en_US slightly favours faker (0.979 vs. 0.983).

4.3 Few-shot regurgitation: a prompting failure, not a quantization failure

A pilot run of the pipeline using a fixed three-shot demonstration template (one English, one Japanese, one Spanish demo) revealed a striking failure mode: across all 509 unique-entity `Bonsai-1.7B` calls, the model produced output that was never literally identical to the input (0 echoes by our validation), but for low-resource-locale inputs the output was very often **one of the few-shot demonstration values verbatim, regardless of the input**:

- 杨娟 (Chinese, NAMED INSURED on a `zh_CN` auto-insurance form) → "Alice Johnson" (the first PERSON demonstration).
- 宁夏回族自治区兰州县山亭陈街B座572990 → "123 Main Street, Boston MA 02101" (the first ADDRESS demonstration).
- `11-Jul-1998` (DD-Mon-YYYY) → `03/15/1985` (MM/DD/YYYY—*both* type and locale wrong).

Table 2: Average per-document latency (seconds, CPU only).

Mode	Detect	Surrogate	Splice + PPL	Total
redact	∼1.0	0.00	∼0.2	1.2
faker	∼1.0	<0.01	∼0.2	1.2
hybrid	∼1.0	∼34.7	∼0.2	35.9

- 山田太郎 (Japanese), Müller-Schulz (German) → both "Alice Johnson".

Diagnosis: prompting, not quantization. Our initial hypothesis was that 1-bit quantization had degraded the model's instruction-following. We tested this by re-running the *exact same five problem prompts* under 1.58-bit `Ternary-Bonsai-1.7B` (Q2_0) [11]—a different quantization scheme running on the same Qwen3 base architecture [12] and the same demonstrations. `Ternary-Bonsai-1.7B` produced **byte-for-byte identical outputs** to `Bonsai-1.7B` (杨娟 → "Alice Johnson", `11-Jul-1998` → `03/15/1985`, etc.) at roughly six times slower per-call inference. This rules out quantization as the root cause: the model is doing what *any* small LM asked to pattern-match `Real:X\nFake:Y` would do—extract the most recent demonstration output as the answer when the input does not match the demonstration's distribution.

Fix: locale-conditioned rotating few-shot demonstrations (described in Section 3.2). With character-range locale detection feeding into pools of 4–8 demonstrations per locale (and 4 demonstrations per date format), and per-input-hash-seeded sampling of 3 demonstrations per call, the same 1-bit `Bonsai-1.7B` produces:

- 杨娟 → 李伟 (Chinese name in zh-pool)
- 宁夏回族自治区兰州县山亭陈街B座572990 → 广东省广州市天河区珠江新城200号 (Chinese address)
- `11-Jul-1998` → `05-Aug-2003` (DD-Mon-YYYY format preserved)
- Müller-Schulz → Anna Becker (German name)
- `Hauptstraße 45, 10117 Berlin` → `Bahnhofstraße 7, 60313 Frankfurt`
- "John Smith" → "David Kim" (different US name; no longer "Alice Johnson")

The fix is dataset-free, does not require fine-tuning, and adds < 50 ms per surrogate-proposal call. All quantitative numbers in Tables 1 and 3 are produced under this fixed prompting strategy.

Known limit: kanji-only Japanese names. These map to the `zh` pool because our locale heuristic requires kana to disambiguate (山田太郎 → 郑强 instead of a Japanese fake). A character-frequency-based classifier or an explicit "treat ambiguous CJK as both ja and zh" strategy would address this.

4.4 Latency

The 30× latency gap between hybrid and redact / faker is dominated by `Bonsai-1.7B` surrogate generation. Without our `(canonical, label)` cache, hybrid would call `Bonsai-1.7B` for every PII mention (∼660 calls across the corpus); with the cache, only **509 unique calls** were made, saving ∼25% of inference time. Each `Bonsai-1.7B` call costs ∼7–10 s for the ∼30-token output (loading + 1.7 B model inference + cleanup). Detection cost is identical across modes because the privacy filter is invoked both for input PII detection and for residual-leak measurement.

Table 3: Span-level NER F1 on held-out original documents (47 train / 11 test). Mean ± SD across 5 spaCy training seeds; the substituted training corpus is fixed per mode, only spaCy gradient initialisation varies.

Mode	Train spans	Precision	Recall	F1	ΔF1 vs. orig.
original	283	0.830 ± 0.033	0.653 ± 0.090	$\mathbf{0.726 \pm 0.046}$	(baseline)
redact	283	0.000 ± 0.000	0.000 ± 0.000	$\mathbf{0.000 \pm 0.000}$	-0.726
faker	283	0.843 ± 0.042	0.310 ± 0.071	$\mathbf{0.448 \pm 0.082}$	-0.278
hybrid	283	0.835 ± 0.061	0.240 ± 0.039	$\mathbf{0.371 \pm 0.049}$	-0.355

4.5 Downstream NER utility

Setup. We test whether substitution preserves *downstream training utility*: an NER model trained on substituted data should approach the F1 of one trained on original data. We use the 58 English-locale documents from the evaluation set (en_US + en_IN), stratified-split 47 train / 11 test by locale. The test set is **always in original form**—substitution applies only to the training set. PII spans are extracted from the ground-truth `pii_gt` dictionary (substring search) so that substitution quality is decoupled from detection quality. A blank `spaCy` English NER pipeline is trained for 30 iterations with a single binary `PII` label on each variant of the train set, then evaluated against held-out original spans (label-agnostic span overlap).

Redact destroys downstream utility entirely (F1 = 0.000). A NER model trained on `[PRIVATE_PERSON]`-style placeholders learns to predict only those placeholder *tokens* and never fires on real text. This is the sharpest possible motivation for substitution: redacted-text training data is, for any downstream NER consumer, effectively *labelled negative-only data*. The model converges (training loss falls to <2.0) but its decision boundary lives in the wrong vocabulary.

Faker and hybrid both recover a substantial fraction of original F1. Type-preserving fakes—whether random (faker) or SLM-generated locale-conditioned (hybrid)—give the model enough surface variety to learn that "capitalised proper-noun-shaped two-token sequence in a name field" is a PII span. Faker recovers $0.448/0.726 = 61.7\%$ of the original baseline; hybrid recovers $0.371/0.726 = 51.1\%$. Precision is high in both cases (0.83–0.84) but recall is only 0.24–0.31—fake-data-trained models are conservative.

Hybrid is plausibly slightly below faker, but the gap is within seed-level noise. The mean F1 difference is $0.448 - 0.371 = 0.077$ in faker's favour. A Welch's two-sample t-test (with $n_f = n_h = 5$ and unequal variances) gives standard error $\mathrm{SE} = \sqrt{0.082^2/5 + 0.049^2/5} = 0.0427$, $t = 0.077/0.0427 = 1.80$, Welch–Satterthwaite degrees of freedom ≈ 6.5, and two-tailed $p \approx 0.12$—not significant at $p < 0.05$. By contrast, the gaps between `original` and either substituted mode, and between any substituted mode and `redact`, are multiple SDs apart and clearly significant.

A pilot run with the naive fixed-three-demonstration strategy from Section 4.3 (single seed) produced hybrid F1 = 0.310 vs. faker F1 = 0.427—a gap of −0.117, larger than what we now observe under the locale-conditioned strategy (−0.077 in mean). Locale-conditioned rotating demonstrations therefore visibly *reduce*, even if they do not fully eliminate, the residual hybrid-below-faker trend within our small-sample regime.

Variance, sample size, and what the seeds reveal. Compared with the baseline F1 noise (original SD = 0.046), faker exhibits the highest seed-to-seed variance (SD = 0.082); hybrid is *tighter* than faker (SD = 0.049). One interpretation is that faker's i.i.d. random surrogates create a more lottery-like training distribution—some seeds happen to land on lucky vocabulary, others do not—whereas hybrid's locale-clustered surrogates produce a more predictable training corpus. Distinguishing this hypothesis from mere $n = 5$ chance would require either a larger held-out test set or nested cross-validation; both are outside the scope of this short paper. The robust within-seed ordering—`original` $>$ `faker` $\gtrsim$ `hybrid` $\gg$ `redact`—holds for every seed we ran (5/5).

5 Discussion and Limitations

Privacy-filter recall ceiling. The non-zero leak rate in all three modes reflects the privacy filter's measured detection F1 of 0.587 (Section 2). A pipeline using a higher-recall detector or pattern-based fallback (e.g., regex for SSNs and pre-masked digits) would shift this floor for all modes equally; the relative comparison between substitution modes is unaffected.

Few-shot regurgitation is a prompting issue, not a quantization issue. Section 4.3 shows that the failure occurs identically under 1-bit `Bonsai-1.7B` and 1.58-bit `Ternary-Bonsai-1.7B`. Larger or higher-precision models *might* suppress the symptom by attending more strongly to the instruction, but the underlying pattern-matching pull will remain at any small-LM scale. The robust fix is at the prompt level (locale-conditioned rotating demonstrations); we recommend it as default practice for any small-LM substitution pipeline.

No coreference resolution. "John" and "John Smith" within the same document are treated as distinct entities by surface-form grouping. A proper coreference layer would further improve consistency.

Naturalness via `distilgpt2` PPL. `distilgpt2` [5] is a weak naturalness proxy: it favours short placeholder tokens (which redact mode produces) over longer realistic surrogates, and is English-only. A proper human evaluation, or PPL under a larger multilingual LM, would refine this measurement.

No explicit threat model. We measure literal-string leak, not inference attacks. A determined adversary with access to (output_text, external knowledge) could potentially link surrogates back to originals. Differential-privacy guarantees on the substitution distribution are out of scope.

Sample size and statistical reporting. The downstream-NER experiment uses 47 train / 11 test documents and $n = 5$ training seeds; we report mean $\pm$ SD per mode and apply Welch's t-test to the two-substituted-mode comparison. The `original` $>$ `faker, hybrid` $\gg$ `redact` ordering is multiple SDs apart for every comparison and significant at $p < 0.001$; only the `faker` vs. `hybrid` gap sits inside the noise band ($p \approx 0.12$). A larger test set, nested cross- validation, or a more powerful significance test (paired-bootstrap over per-seed F1 differences) would tighten this last comparison.

6 Conclusion

We built and evaluated a fully on-device PII substitution pipeline combining a token classifier (`openai/privacy-filter`), a 1-bit small language model (`Bonsai-1.7B`, Q1_0), and a rule-based generator (`faker`). The architecture works as intended at the document level: privacy is preserved (within the privacy-filter's recall ceiling), surrogates are consistent across mentions, length preservation is best-of-three across five of six locales, and naturalness PPL beats faker-only baselines across all six locales.

The most actionable contribution is methodological. We document a "few-shot regurgitation" failure mode in which a small SLM, prompted naively with a fixed three-shot demonstration template, ignores the input and emits one of the demonstration *outputs* verbatim—and we show by direct comparison with 1.58-bit `Ternary-Bonsai-1.7B` that this is a property of *prompting* at small-LM scale, not of 1-bit quantization. A simple locale-conditioned rotating-demonstrations prompting strategy fixes it: the fixed pipeline produces locale-correct, format-preserving surrogates (杨娟 → 李伟, Müller-Schulz → Anna Becker, `11-Jul-1998` → `05-Aug-2003`), eliminates the previously observed regression in downstream NER training F1, and adds < 50 ms per surrogate proposal.

Substitution outperforms `[LABEL]`-style redaction by an enormous margin on downstream NER training utility (F1 0.371 ± 0.049 vs. 0.000 ± 0.000 across 5 seeds; the gap is many standard deviations, not a noise effect). Whether SLM-generated surrogates can *exceed* random faker surrogates on downstream utility—rather than match them within seed-level noise—remains an open question for a longer evaluation with a larger test set or nested cross-validation, which we believe is the most direct way to push the hybrid–faker comparison out of the $p \approx 0.12$ band we observe at $n = 5$.

All code, configurations, and the 100-document evaluation harness are released under an open license.

Reproducibility

Code, configurations, and the 100-document evaluation set (`samples.json`, the synthetic multi-template benchmark shipped with the `openai/privacy-filter` inference harness) used in this paper are released at
`https://github.com/asadani/on-device-pii-substitution`. To reproduce the primary results:

```
# Primary metrics (Table 1, ~60 minutes on CPU)
python eval_substitution.py --n 100 --modes redact faker hybrid \
       --bonsai-size 1.7B --output results/

# Per-locale and per-template breakdowns
python analyze_results.py \
       --results results/substitution_results.json \
       --out results/analysis.md

# Downstream NER utility (Table 2, ~25 minutes)
python eval_ner_utility.py --output results/
```

Total wall-clock for the full pipeline is approximately 90 minutes on a 31 GB single-CPU machine. The privacy-filter model (∼2.8 GB) is downloaded once from Hugging Face on first run and cached locally.

Use of generative AI assistance

In line with arXiv's policy on the use of generative AI in scholarly work, the author discloses that a large language model assistant (Anthropic's Claude) was used during the preparation of this manuscript. Specifically: (i) the assistant helped explore and draft prose for the introduction, related-work, and discussion sections from the author's notes and bullet-point outlines; (ii) the assistant helped scaffold LaTeX, bibliography, and table formatting; and (iii) the assistant ran the experimental scripts and reported the resulting numerical outputs back to the author. All quantitative results in Tables 1, 2, and 3 are computed by the released code (`eval_substitution.py`, `eval_ner_utility.py`) on the released artefact and are reproducible by any reader; no result was generated, projected, or hallucinated by the language model. The author is solely responsible for the technical claims, methodology, statistical analysis, and the final wording of the manuscript.

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A Locale-conditioned demonstration pools

For full reproducibility we list the demonstration pools used by the locale-conditioned rotating few-shot strategy (Section 3.2). Per-input MD5 hashes seed a deterministic sample of three demonstrations from the appropriate pool.

PERSON pool.

- `en`: (John Carter, Marcus Chen), (Linda Vasquez, Olivia Brennan), (David Kim, Theo Pemberton), (Sarah Patel, Maya Iyer), (Robert Williams, Daniel Foster), (Priya Krishnamurthy, Nadia Subramanian), (Michael O'Brien, Patrick Donovan), (Jennifer Wong, Cynthia Park).
- `de`: (Hans Müller, Karl Schmidt), (Anna Becker, Lena Hoffmann), (Klaus Wagner, Erik Krüger), (Ingrid Weber, Petra Neumann), (Stefan Fischer, Dietrich Bauer), (Helga Zimmermann, Brigitte Klein).
- `es`: (Juan García, Carlos Hernández), (María Rodríguez, Ana Fernández), (Diego Sánchez, Luis Castillo), (Carmen Ortiz, Lucía Vázquez), (Roberto Jiménez, Pablo Morales), (Sofía Ramírez, Elena Aguilar).
- `ja`: (山田太郎, 鈴木一郎), (佐藤花子, 田中美咲), (渡辺健, 高橋翔), (中村裕子, 小林由香), (加藤博之, 斎藤大輔), (井上恵美, 松本香織).
- `zh`: (李伟, 王芳), (张敏, 刘洋), (陈杰, 黄燕), (周磊, 吴娟), (徐明, 孙丽), (郑强, 马晶).

ADDRESS pool. Six entries per locale; en covers both en_US and en_IN. Examples include (`Hauptstraße 45, 10117 Berlin`, `Lindenallee 12, 80331 München`) for `de`, (`Calle Reforma 123, 06600 CDMX`, `Avenida Insurgentes 456, 03100 CDMX`) for `es`, and (北京市朝阳区建国路1号, 上海市浦东新区世纪大道100号) for `zh`. Full lists are in `pii_substitute.py`.

DATE pool. Pools per detected format: `mdy_slash` (5 entries), `ymd_dash` (4), `dmy_dash_mon` (4), `dmy_slash` (3), `unknown` (2 fallbacks).